

DAY 4 – COMPONENT-SPECIFIC TROUBLESHOOTING

Course: Diploma in Cyber Security

**Topic: Component-Specific
Troubleshooting**

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TASK 1 – RAM RESEATING

Practical Steps:

1. Sabse pehle PC ko shut down karunga.
2. Power cable ko disconnect karunga.
3. PC case open karunga.
4. Motherboard par RAM slots find karunga.
5. RAM ke side ke clips ko press karke RAM ko carefully remove karunga.
6. RAM ko check karunga ki koi visible damage ya dust to nahi hai.
7. RAM ko wapas slot mein properly lagaoonga.
8. Clips lock hone ke baad PC case close karunga.

9. Power connect karke PC start karunga.

10. Check karunga ki PC normally start ho raha hai ya nahi.

Observation:

Before: RAM ko remove karne se pehle PC normal working condition mein tha.

After: RAM ko reseal karne ke baad PC start karke display check kiya.

TASK 2 – GPU TROUBLESHOOTING CHECKLIST

GPU Troubleshooting Checklist

1. Check the Display

First, I would check if the monitor is working and showing any display.

2. Check the HDMI/DisplayPort Cable

I would check both ends of the cable and make sure it is connected properly.

3. Check the GPU Connection

I would check if the GPU is properly fitted in the PCIe slot.

4. Check GPU Power Cable

I would check that the GPU power connector is properly connected.

5. Check the Graphics Driver

I would check the graphics driver for any error or problem.

6. Test the PC Again

After checking these things, I would start the PC and check if the display is working normally.

Result:

If the display starts working normally, the problem may have been related to the cable, connection or driver.

TASK 3 – STORAGE ISSUE

Practical Task:

In this task, I simulated a storage drive failure by physically unplugging and reconnecting the primary SATA/NVMe SSD on a test system.

Steps I Followed:

1. Shut down the computer completely and unplugged the power supply.
2. Opened the side panel of the PC case to locate the storage drive (SATA SSD / NVMe M.2 slot).
3. Disconnected the SATA data and power cables (or unseated the M.2 NVMe SSD).
4. Powered on the computer and entered the BIOS/UEFI setup by pressing the DEL / F2 key.
5. Observed the system behavior and boot error screens.

6. Powered off the PC, reconnected the storage drive firmly, and booted up again.

Observations:

- **Before Reconnecting (Drive Disconnected):**

- **BIOS Screen:** Under "Storage Information / Boot Priority", the drive status showed None / Not Detected.
- **System Error Message:** The PC failed to load the operating system and displayed:

"Reboot and Select proper Boot device or Insert Boot Media in selected Boot device and press a key" (or "No Bootable Device Found").

- **After Reconnecting (Drive Plugged In):**

- **BIOS Screen:** The storage drive model name and capacity were immediately detected under SATA/NVMe configuration.
- **System Status:** Windows OS booted normally to the login screen without any errors.

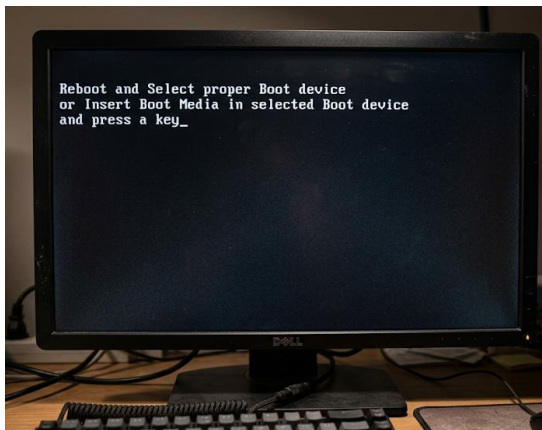


Figure 2: Storage drive disconnection simulation and BIOS boot device detection.

TASK 4 – RANDOM RESTART DIAGNOSTIC SCENARIO

Problem Scenario:

A PC is restarting randomly during normal use. The candidate components to diagnose are RAM, GPU, and Storage.

Which Component I Would Check First:

I would check the RAM first.

Reasoning:

A faulty RAM module, loose memory contact, or bad memory sectors directly cause unexpected memory address exceptions, kernel panics (BSOD), and immediate system restarts. Unlike storage failure (which usually freezes the system or throws boot errors) or GPU failure (which typically causes display artifacts or black screens), unstable RAM immediately interrupts CPU execution, forcing the PC to reboot randomly.

Diagnostic Steps for RAM:

1. Physical Inspection & Reseating: Power off the PC, remove both RAM sticks, clean the gold contact pins with a dry microfiber cloth/eraser, and firmly reseat them into slots A2 and B2 until they click.

2. Single-Stick Isolation Test: If multiple RAM sticks are installed, test the PC using one RAM stick at a time to identify if a specific module is faulty.
3. Slot Testing: Test a single known-good RAM stick across different motherboard DIMM slots to rule out a damaged memory slot.
4. Memory Diagnostic Utility: Boot the system into Windows and run the built-in Windows Memory Diagnostic Tool (mdsched.exe) or boot from a MemTest86 USB drive to scan for memory read/write hardware errors.
5. Secondary Checks (If Issue Persists): If the RAM passes all tests with zero errors, proceed to check GPU temperatures (overheating) and Storage drive health using SMART status tools (CrystalDiskInfo).

CONCLUSION

In this practical troubleshooting session, I learned how to systematically isolate and resolve common hardware faults involving RAM, GPU, and Storage drives. I practiced the physical reseating process for memory modules and understood how dirty or loose pins cause system instability. I developed a structured checklist to troubleshoot GPU display failures from physical cabling up to driver integrity. Furthermore, through the storage disconnection simulation, I observed the exact BIOS indicators and "No Boot Device"

error messages associated with storage drive detection issues. Finally, I applied logical diagnostic reasoning to prioritize RAM testing for random reboot scenarios, reinforcing the importance of step-by-step troubleshooting before replacing hardware.